

Quadratic functions - Set 3 - Solutions

Specification MM1.3, M4.5, M4.11-M4.14.

Answer key

Question	Answer	Topic
1	D	Quadratic functions
2	H	Quadratic functions

Worked solutions

Question 1 QUA-22 · Quadratic functions · spec MM1.3

The equation $x^2 + ax + b = 0$ has roots 3 and -5 . The equation $x^2 + bx + a = 0$ has roots p and q . Find $p^2 + q^2$.

Answer: D

Sum of roots $3 + (-5) = -a$, so $a = 2$; product $-15 = b$. Then $x^2 - 15x + 2 = 0$ has $p + q = 15$ and $pq = 2$, so $p^2 + q^2 = 15^2 - 2(2) = 221$.

Full method - how and why

Two linked quadratics: the roots of the first give the coefficients of the second. Sum-and-product relations do all the work.

- 1 For $x^2 + ax + b = 0$ with roots 3 and -5 : $\text{sum} = 3 + (-5) = -2 = -a$, so $a = 2$.

why: Sum of roots $= -b/a$; with a monic quadratic that is just $-a$ - mind the double negative.

- 2 Product $= 3 \times (-5) = -15 = b$.

why: Product of roots $= c/a = b$ here.

- 3 The second equation is $x^2 - 15x + 2 = 0$, so $p + q = 15$ and $pq = 2$.

why: Substitute the values found, being careful that the roles of a and b swap between the two equations.

- 4 Use $p^2 + q^2 = (p + q)^2 - 2pq = 225 - 4 = 221$.

why: The same identity as in QUA-04.

Where it goes wrong

- Getting $a = -2$ by forgetting the minus in 'sum $= -a$ ', which turns the second equation into $x^2 - 15x - 2$ and gives 229.
- Computing $(p + q)^2$ and forgetting to subtract $2pq$, giving 225.

Takeaway Read coefficients from roots with $\text{sum} = -b/a$ and $\text{product} = c/a$, and keep track of which equation you are in.

Question 2 X2-01 · Quadratic functions · spec MM1.3

p and q are real numbers. Which of the following statements is/are true?

- I If $p^2 < 4q$ then $x^2 + px + q > 0$ for all real x .
- II If $x^2 + px + q > 0$ for all real x then $q > 0$.
- III If $q < 0$ then $x^2 + px + q = 0$ has two real roots of opposite sign.

Answer: H

I: negative discriminant and positive leading coefficient. II: put $x = 0$. III: the product of the roots is $q < 0$ (and the discriminant $p^2 - 4q > 0$), so the roots are real with opposite signs.

Full method - how and why

Three claims about a quadratic. Each is settled by the discriminant, by a well-chosen substitution, or by the root-product relationship.

- 1 I: $p^2 < 4q$ means the discriminant $p^2 - 4q < 0$, so $x^2 + px + q$ has no real roots; with a positive x^2 coefficient the graph lies entirely above the axis. **True.**
why: No roots plus an upward parabola means always positive - the graph never reaches the axis.
- 2 II: if $x^2 + px + q > 0$ for **all** x , then in particular at $x = 0$: $q > 0$. **True.**
why: Substituting a convenient value is the quickest way to extract information from a universal statement.
- 3 III: the product of the roots is q . If $q < 0$ then $p^2 - 4q > p^2 \geq 0$, so the roots are real and distinct, and their product being negative makes them opposite in sign. **True.**
why: For $x^2 + px + q$ the sum of roots is $-p$ and the product is q .
- 4 So all three.
why: Match to the list.

Where it goes wrong

- Reading I's condition as $p^2 > 4q$ - that would give two real roots and the opposite conclusion.
- Forgetting to check in III that the roots are **real**; a negative product alone would be meaningless for complex roots.
- Trying to prove II with a general argument when $x = 0$ settles it in one line.

Takeaway Discriminant controls the number of roots; $x = 0$ extracts the constant term; sum and product of roots are $-b/a$ and c/a .